

Mediterranean diet and non enzymatic antioxidant capacity in the PREDIMED study: evidence for a mechanism of antioxidant tuning.

BACKGROUND AND AIMS: The intake of antioxidant-rich foods may increase the blood levels of non enzymatic antioxidant capacity (NEAC). NEAC takes into account all antioxidants from food and synergistic effects between them. We examined the effect of a 1-year intervention with Mediterranean diet on plasma NEAC and assessed whether it was related to baseline NEAC levels. **METHODS AND RESULTS:** Five hundred sixty-four participants at high cardiovascular risk were randomly selected from the PREDIMED (Prevención con Dieta MEDiterránea) Study, a large 3-arm randomized clinical trial. Blood NEAC levels were measured at baseline and after 1-year of dietary intervention with 1) a Mediterranean diet supplemented with virgin olive oil (MED + VOO); 2) a Mediterranean diet supplemented with nuts (MED + nuts), or 3) a control low-fat diet. Plasma NEAC was analyzed using FRAP (ferric reducing antioxidant potential) and TRAP (total radical-trapping antioxidant parameter) assays. Plasma FRAP levels increased after 1-year of intervention with MED + VOO [72.0 $\mu\text{mol/L}$ (95% CI, 34.2-109.9)] and MED + nuts [48.9 $\mu\text{mol/L}$ (24.3-73.5)], but not after the control low-fat diet [13.9 $\mu\text{mol/L}$ (-11.9 to 39.8)]. Participants in the lowest quartile of plasma FRAP at baseline significantly increased their levels after any intervention, while those in the highest quartile decreased. Similar results occurred with TRAP levels. **CONCLUSIONS:** This study shows that a 1-year of MED diet intervention increases plasma TAC level in subjects at high risk for cardiovascular disease. Moreover, the effectiveness of dietary supplementation with antioxidants may be related to baseline levels of plasma NEAC. © 2013 Elsevier B.V. All rights reserved.